

# Natural Language Processing VU (DAT.C309UF)

## SS26

### Practicals - Default Project

April 2026

Release Data: 22.04.2026

Release Test Data: 18.06.2026

Deadline Stage 1, Stage 2 and report: 25.06.2026

**Submission: via TeachCenter**

**Group Size: 2**

**Attention: Please adhere to all formatting and length requirements!**

**Do not forget to begin early, especially Stage 2 takes a lot of computing time!**

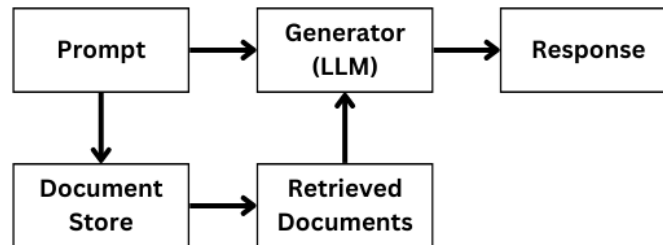
## 1 Introduction

The goal of this practical is to get you to not only know a few NLP methods but to also apply them. You will have to use these to determine if a given text is either an opinion or a fact. In addition to this you will also utilize knowledge defined by you beforehand to help classify these texts. Therefore, you also have to know about **knowledge bases** and **retrieval augmented generation**:

A **knowledge base** is a centralized repository designed to store, organize, and manage information systematically. This enables users or systems to retrieve and use knowledge efficiently. It acts as a database but is typically more focused on facilitating the understanding and automated processing of its stored data. A more complex example is the [gene ontology](#), a knowledge base on the functions of genes. Think about how this can help you classify texts into either opinion or fact.

**Retrieval augmented generation (RAG)** uses external knowledge sources to inform and guide the text generation process of a Large language model (LLM). This approach allows for new information to be integrated into the model's responses without having to retrain the whole model. The following flowchart shows a simplified version of this process:

## Retrieval Augmented Generation



In order to narrow down the topic you will have to build your knowledge base and RAG approach around the text documents you will have to classify.

### 1.1 Submission

*Because of the delayed release of the dataset and the assignment description, we decided to merge both stage 1 and stage 2. This gives you more freedom in solving each stage.*

This means you must complete **both stages** by **25.06.2026**. We will release a validation set to test your implementation 1 week before the deadline on the 18.06.2026. While this is primarily to test your final pipeline on unseen data, we also expect you to upload the classification labels using both stages. You will not be graded on how well or poorly your approach works, but rather on the overall pipeline and methodology used.

Altogether, you have to hand in:

- Report: `groupXX_report.pdf`
- Submission: `groupXX_submission.zip` including your code and classification results of both stages, `groupXX_classifications_1.csv` for stage 1 and `groupXX_classifications_2.csv` for stage 2.

For all your submissions, **please adhere to the naming and formatting convention above**. You can find an example submission with dummy data on the TC, as well as a template for the report. Please use this for your submissions. In your report, do not exceed the **6 page limit**. Both stages must be clearly separated in both the report and code, even though some overlap is, of course, possible. If these criteria are not met or if the CSV file is not parsable, we will deduct points.

Lastly, please note that we do not condone copy-pasting solutions or approaches from LLM. At the end of June, we will hold assignment interviews. The details will be announced on Teachcenter.

## 1.2 Grading

- **Projects** (60 points total)
  - **Data Verification** (up to -10 penalty if failed): Authenticity of data-points (source links) and label correctness.
  - **Stage 1 Submission** (20 points): Problem-solving approach, solution complexity, and evaluation (must include a baseline).
  - **Stage 2 Submission** (20 points): Problem-solving approach, solution complexity, and evaluation (must include a baseline).
  - **Final Report** (15 points): Clarity and quality of the presentation of the problem and solution.
  - **Source Code** (5 points): Quality, portability, and reproducibility.

## Grading Scheme

The final grade is determined by the total accumulated points  $P$  out of 80: Please note that the missing 20 points stem from the individual homework.

| Point Range Grade  |   |
|--------------------|---|
| $0 \leq P \leq 40$ | 5 |
| $40 < P \leq 50$   | 4 |
| $50 < P \leq 60$   | 3 |
| $60 < P \leq 70$   | 2 |
| $70 < P \leq 80$   | 1 |

## 2 Stages

### 2.1 Stage 1 - Traditional NLP

For the first stage of the project we will provide you with a dataset that has labels assigned to each entry. We want you to employ traditional NLP techniques to classify the text documents. Additionally, we want you to develop a simple knowledge base that helps you determine if a given text is an opinion or a fact. Then use the features extracted to train a basic machine learning model to classify articles as 'Opinion' or 'Fact'. Think about what differentiates an opinion from a fact and how you can extract this from a given text.

**Do not use transformers for this task!**

#### Hints for Stage 1

Some linguistic features you could take a look at are:

- **POS Tagging:** Analyze the part of speech of words in the articles. Pay attention to the usage of nouns, verbs, and adjectives, their frequency and arrangement.
- **Named Entity Recognition (NER):** Identify and classify proper nouns in the text, such as names of people, organizations, or locations.
- **Emotion Analysis:** Determine the emotional tone of the articles.
- **Readability Analysis:** Evaluate the complexity of the text. Analyze sentence structure and word difficulty.
- **Error Analysis:** Identify grammatical and spelling errors.

Also, consider what preprocessing steps you need for each of your subtasks.

### 2.2 Stage 2 - LLM-Based Text Classification

In the last task, you will classify text documents using a LLM. For this, you will need to expand your knowledge base. **We do not give you strict guidelines** dictating exactly how you must achieve this. Instead, you are entirely free to be creative and explore the capabilities of modern LLMs. You have the freedom to design your own pipeline and methodology.

**Core Requirement:** You must select an approach, implement it, and comprehensively **explain why your chosen method is useful and effective** for the task of distinguishing between facts and opinions. Consider the linguistic and contextual challenges of this specific classification when writing your justification.

## Possible Approaches

You are free to choose from, combine, or expand upon the following methodologies:

- **Zero-Shot Learning:** Instructing the LLM to classify the text directly via a carefully engineered system prompt without providing any prior examples. This can serve as a baseline but is not enough for stage 2.
- **One-Shot / Few-Shot Learning:** Providing the LLM with one or several carefully curated examples of facts and opinions within the prompt context to guide its reasoning pattern.
- **Chain-of-Thought (CoT) Prompting:** Forcing the model to explain its reasoning step-by-step before outputting the final classification, which is particularly useful for untangling complex, mixed statements.
- **Retrieval-Augmented Generation (RAG):** Using an external knowledge base or search tool to verify claims and retrieve context about the text before asking the LLM to classify it.
- **Fine-Tuning:** Updating the model’s weights using a labeled dataset of facts and opinions. You may explore full fine-tuning or Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA or QLoRA.
- **Self-Consistency / Multi-Agent Debate:** Generating multiple reasoning paths or using multiple LLM instances to debate the classification of a document and voting on a final output.
- **Prompt Tuning / Soft Prompting:** Optimizing a continuous vector representation of the prompt through gradient descent to adapt the LLM specifically for opinion-fact differentiation.

No matter what approach you tried, tell us about it in the report, including a quick overview of why you stopped trying a specific approach and decided on a different one. You can e.g. present inferior results produced by one approach that another approach improved on.

You can use any LLM you like for this task and using a less advanced LLM and thus getting worse results will not impact the points you will receive. Some possible LLMs to use are LLMs of the Llama family, older GPT models like GPT-2, or BART.